

| Tool Version : Vivado v.2025.1 (win64) Build 6140274
Thu May 22 00:12:29 MDT 2025
| Date : Mon Apr 13 17:37:14 2026
| Host : DESKTOP-STEMT34 running 64-bit major
release (build 9200)
| Command : report_clock_utilization -file
top_clock_utilization_routed.rpt
| Design : top
| Device : 7a100t-csg324
| Speed File : -1 PRODUCTION 1.23 2018-06-13
Design State : Routed

Clock Utilization Report

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1. Clock Primitive Utilization

+-----+-----+-----+-----+-----+-----							
----+							
Type	Used	Available	LOC	Clock Region			
Pblock							
+-----+-----+-----+-----+-----+-----							
----+							
BUFGCTRL	2	32	0	0			
0							
BUFH	0	96	0	0			
0							
BUFIO	0	24	0	0			
0							
BUFMR	0	12	0	0			
0							
BUFR	0	24	0	0			
0							
MMCM	0	6	0	0			
0							
PLL	0	6	0	0			
0							
+-----+-----+-----+-----+-----+-----							
----+							

2. Global Clock Resources

+-----+-----+-----+-----+				
-----+-----+-----+-----				
-----+-----+-----+-----				
+-----+-----+-----+-----+				
Global Id	Source Id	Driver Type/Pin	Constraint	
Site	Clock Region	Load Clock Region		
Clock Loads	Non-Clock Loads	Clock Period	Clock	
Driver Pin		Net		
+-----+-----+-----+-----+				
-----+-----+-----+-----				
-----+-----+-----+-----				
+-----+-----+-----+-----+				
g0	src0	BUFG/O	None	
BUFGCTRL_X0Y0	n/a		1	
548	0			
cpu_clk_BUFG_inst/O		cpu_clk_BUFG		
g1	src1	BUFG/O	None	
BUFGCTRL_X0Y16	n/a		1	
23	0	10.000	sys_clk_pin	
CLK100MHZ_IBUF_BUFG_inst/O		CLK100MHZ_IBUF_BUFG		
+-----+-----+-----+-----+				
-----+-----+-----+-----				
-----+-----+-----+-----				
+-----+-----+-----+-----+				

* Clock Loads column represents cell count of net connects that connect to a clock pin. Internal cell leaf

* Clock Loads column represents cell count of net connects that connect to a clock pin. Internal cell leaf pins are not considered

** Non-Clock Loads column represents cell count of non-clock pin loads

4. Clock Regions: Key Resource Utilization

Handwriting practice lines with dashed lines and plus signs for tracing.

		Global Clock		BUFRs	
BUFMRs		BUFIOs		MMCM	PLL
GT		PCI		ILOGIC	OLOGIC
FF		LUTM		RAMB18	
RAMB36		DSP48E2			

+-----+-----+-----+-----+-----
 -+-----+-----+-----+-----+-----+-----+-----+-----+-----
 -----+-----+-----+-----+-----+-----+-----+-----+-----
 -----+-----+-----+-----+-----+-----+-----+-----+-----+-----
 -----+-----+-----+-----+-----+-----+-----+-----+-----+-----

	Clock	Region	Name		Used		Avail		Used		Avail		Used	
	Avail		Used		Avail		Used		Avail		Used		Avail	
Used		Avail		Used		Avail		Used		Avail		Used		
Avail		Used		Avail		Used		Avail		Used		Avail		
Used		Avail		Used		Avail								

-----+-----+-----+-----+-----+-----+-----
-+-----+-----+-----+-----+-----+-----+-----+-----
-----+-----+-----+-----+-----+-----+-----+-----
--+-----+-----+-----+-----+-----+-----+-----+-----
-----+-----+-----+-----+

X0Y0				0 12 0 4 0									
2		0		4		0		1		0		1	
0		0		0		0		50		0		50	
0		2600		0		600		0		20		0	
10		0		20									
X1Y0				0 12 0 0 0									
0		0		0		0		0		0		0	
0		4		0		0		0		0		0	
0		1500		0		550		0		40		0	
20		0		40									
X0Y1				1 12 0 4 0									
2		0		4		0		1		0		1	
0		0		0		0		50		0		50	
23		2000		0		600		0		20		0	
10		0		20									
X1Y1				0 12 0 4 0									
2		0		4		0		1		0		1	
0		0		0		0		50		0		50	
0		1900		0		650		0		60		0	
30		0		40									
X0Y2				0 12 0 4 0									
2		0		4		0		1		0		1	
0		0		0		0		50		0		50	
0		2000		0		600		0		20		0	
10		0		20									
X1Y2				1 12 0 4 0									

	2	0	4	0	1	0	1
0	0	0	0	0	50	0	50
516	1900	147	650	0	60	0	
30	0	40					
X0Y3			0	12	0	4	0
	2	0	4	0	1	0	1
0	0	0	0	0	50	0	50
	0	2600	0	600	0	20	0
10	0	20					
X1Y3			0	12	0	0	0
	0	0	0	0	0	0	0
0	4	0	1	0	0	0	0
	0	1350	0	500	0	30	0
15	0	40					
+-----+-----+-----+-----+-----+-----+							
-+-----+-----+-----+-----+-----+-----+-----+							
-----+-----+-----+-----+-----+-----+-----+							
--+-----+-----+-----+-----+-----+-----+-----+							
-----+-----+-----+							

* Global Clock column represents track count; while other columns represents cell counts

5. Clock Regions : Global Clock Summary

All Modules

	X0	X1
Y3	0	0

Y2	0	0
Y1	0	0
Y0	0	0

6. Device Cell Placement Summary for Global Clock g0

Global Id	Driver Type/Pin	Driver Region (D)	Clock	Period (ns)	Waveform (ns)	Slice Loads	IO Loads	Clocking Loads	GT Loads	Net
g0	BUFG/O	n/a				516	0	0	0	cpu_clk_BUFG

* Slice Loads column represents load cell count of all cell types other than IO, GT and clock resources

** IO Loads column represents load cell count of IO types

*** Clocking Loads column represents load cell count that are clock resources (global clock buffer, MMCM, PLL, etc)

**** GT Loads column represents load cell count of GT types

	X0	X1	HORIZONTAL PROG DELAY
Y3	0	0	-
Y2	0	516	0
Y1	0	0	-
Y0	0	0	-

7. Device Cell Placement Summary for Global Clock g1

Global Id	Driver Type/Pin	Driver Region (D)	Clock	Period (ns)	Waveform (ns)	Slice Loads	IO Loads	Clocking Loads	GT Loads	Net
g1	BUFG/O	n/a	sys_clk_pin	10.000	{0.000 5.000}					23

```

|          0 |          0 |          0 |
CLK100MHZ_IBUF_BUFG |
+-----+-----+-----+-----+
-----+-----+-----+-----+
-----+-----+-----+-----+
-+

```

	X0	X1	HORIZONTAL PROG DELAY
Y3	0	0	-
Y2	0	0	-
Y1	23	0	0
Y0	0	0	-

```
8. Clock Region Cell Placement per Global Clock: Region
X0Y1
```



```
# Location of IO Primitives which is load of clock spine

# Location of clock ports
set_property LOC IOB_X1Y126 [get_ports CLK100MHZ]

# Clock net "cpu_clk_BUFG" driven by instance
"cpu_clk_BUFG_inst" located at site "BUFGCTRL_X0Y0"
#startgroup
create_pblock {CLKAG_cpu_clk_BUFG}
add_cells_to_pblock [get_pblocks {CLKAG_cpu_clk_BUFG}]
[get_cells -filter { PRIMITIVE_GROUP != I/O &&
IS_PRIMITIVE==1 && PRIMITIVE_LEVEL !=INTERNAL }
-of_object [get_pins -filter {DIRECTION==IN} -of_objects
[get_nets -hierarchical -filter
{PARENT=="cpu_clk_BUFG"}]]]
resize_pblock [get_pblocks {CLKAG_cpu_clk_BUFG}] -add
{CLOCKREGION_X1Y2:CLOCKREGION_X1Y2}
#endgroup

# Clock net "CLK100MHZ_IBUF_BUFG" driven by instance
"CLK100MHZ_IBUF_BUFG_inst" located at site
"BUFGCTRL_X0Y16"
#startgroup
create_pblock {CLKAG_CLK100MHZ_IBUF_BUFG}
add_cells_to_pblock [get_pblocks
{CLKAG_CLK100MHZ_IBUF_BUFG}] [get_cells -filter {
PRIMITIVE_GROUP != I/O && IS_PRIMITIVE==1 &&
PRIMITIVE_LEVEL !=INTERNAL } -of_object [get_pins
-filter {DIRECTION==IN} -of_objects [get_nets
-hierarchical -filter {PARENT=="CLK100MHZ_IBUF_BUFG"}]]]
resize_pblock [get_pblocks {CLKAG_CLK100MHZ_IBUF_BUFG}]
```

```
-add {CLOCKREGION_X0Y1:CLOCKREGION_X0Y1}  
#endgroup
```